

Problem 1 – Question 2

Axis-angle representation of an arbitrary $U(2)$ matrix

Let $U \in U(2)$. Since $\det U$ has unit modulus, choose

$$\alpha = \frac{1}{2} \arg(\det U)$$

and define

$$V = e^{-i\alpha} U.$$

With a consistent choice of the square root of $\det U$, one has $\det V = 1$, so $V \in SU(2)$. Every $SU(2)$ matrix has the form

$$V = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1.$$

Write

$$a = a_R + ia_I, \quad b = b_R + ib_I.$$

Then

$$V = a_R \mathbb{I} + i(b_I X + b_R Y + a_I Z).$$

The coefficient vector is real, and the normalization condition becomes

$$a_R^2 + a_I^2 + b_R^2 + b_I^2 = 1.$$

Choose θ so that

$$\cos\left(\frac{\theta}{2}\right) = a_R, \quad s = \sin\left(\frac{\theta}{2}\right) = \sqrt{a_I^2 + b_R^2 + b_I^2}.$$

If $s \neq 0$, define the real unit vector

$$\mathbf{n} = -\frac{1}{s} (b_I, b_R, a_I).$$

Using the result of Question 1,

$$V = \cos\left(\frac{\theta}{2}\right) \mathbb{I} - i \sin\left(\frac{\theta}{2}\right) \mathbf{n} \cdot \boldsymbol{\sigma} = R_{\mathbf{n}}(\theta).$$

Therefore

$$\boxed{U = e^{i\alpha} R_{\mathbf{n}}(\theta)}.$$

When $s = 0$, $V = \mathbb{I}$ or $V = -\mathbb{I}$. These cases may be represented by $\theta = 0$ or $\theta = 2\pi$, respectively, with an arbitrary unit axis $\mathbf{n}$. The parameters are not unique: changing the branch of α , replacing $(\mathbf{n}, \theta)$ by $(-\mathbf{n}, -\theta)$, and using the 4π -periodicity of spin rotations give equivalent representations.

Coordinate-free parameter extraction. For $s \neq 0$, the same result can be written as

$$\cos\left(\frac{\theta}{2}\right) = \frac{1}{2} \text{Tr } V, \quad n_j = \frac{i}{2s} \text{Tr}(\sigma_j V), \quad j \in \{X, Y, Z\}.$$

For $V \in SU(2)$, these quantities are real.